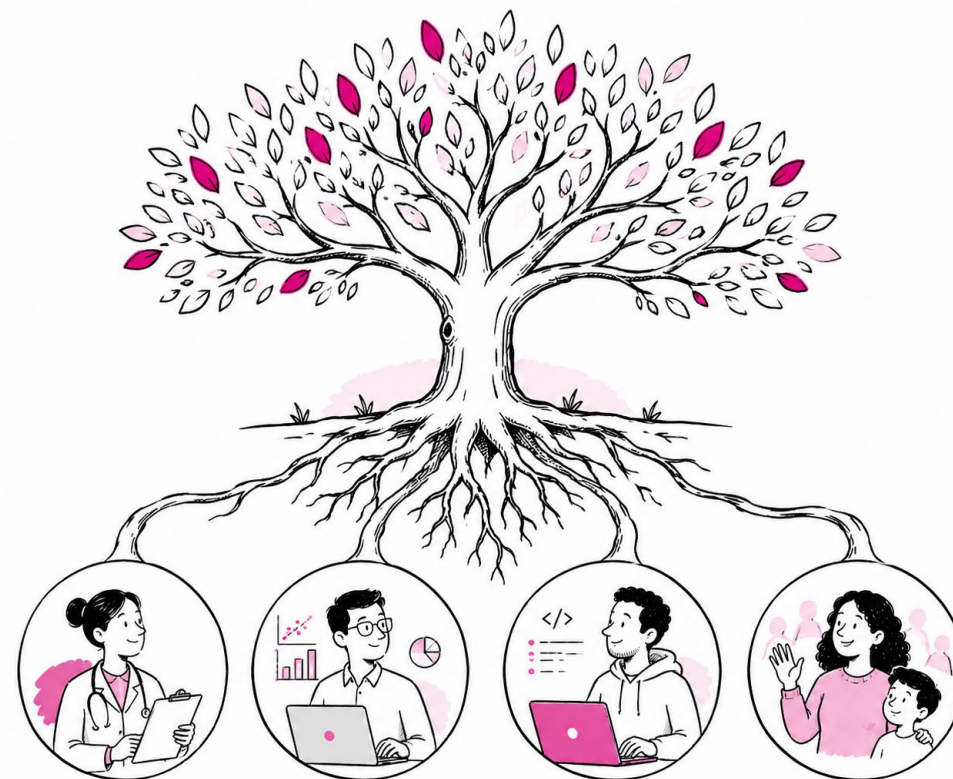


BETTER TEAMS, BETTER SCIENCE

*Making collaborative change in
biomedical data science*

**Biomedical Data Science
Leadership Alliance**



Welcome



Charlotte George

INTEGRATE, Co-Project Lead
and Genomic Data Science
Trainer, University of Oxford



David Sims

INTEGRATE, Co-Project Lead
and Associate Professor of
Computational Genomics,
University of Oxford

Agenda for the day

Time	Activity
09:30-10:00	Arrival, registration and networking breakfast
10:00-10:40	Welcome Introductory talk from Biomedical Data Science Leadership Alliance (BDSLA)
10:40 – 11:20	Panel discussion: ‘Future Directions and Drivers for Change in Biomedical Data Science and Team Research’
11:20 – 11:35	Coffee break
11:35 – 12:35	Activity: Making change happen Part 1: Visioning and current activities
12:35 – 13:30	Lunch break
13:30 – 15:00	Activity: Making change happen Part 2: Defining actionable steps
15:00 – 16:00	Share out , next steps and closing

House keeping

Fire safety

- Fire alarm is not expected.
- If alarm sounds, use nearest fire exit, follow venue staff.
- Meet outside in the Quad.

Comfort breaks and refreshments

- Toilets located on all floors, accessible toilet near main staircase or basement.
- Coffee breaks and lunch in rooms 8 and 9
- Afternoon activity, please grab a coffee at the end, if needed, or after close of event.

Code of conduct

- Please read and adhere to Turing events [CoC](#).
- Facilitators are asked to actively include all, orally or in writing.
- Report to Emma Karoune (ekaroune@turing.ac.uk) or Eirini Koutsouroupa (ekoutsouroupa@turing.ac.uk)

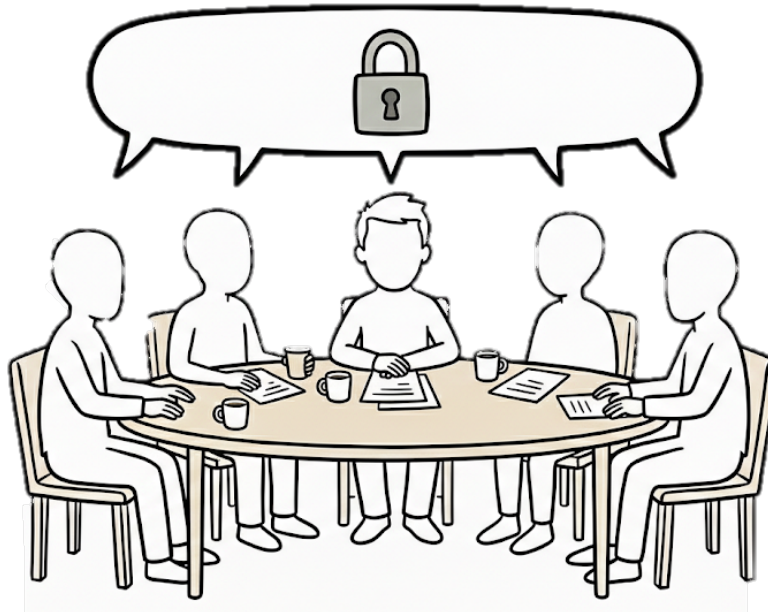
Quiet space

- Please use rooms 8 and 9
- Quad.

To drive open discussions

Chatham House Rule

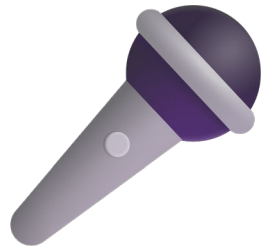
“When a meeting, or part thereof, is held under the Chatham House Rule, participants are free to use the information received, but neither the identity nor the affiliation of the speaker(s), nor that of any other participant, may be revealed.”



Questions during the day

Room:

Hands up and microphone



All:

www.slido.com

#4049254



Presentation on behalf of Biomedical Data Science Leadership Alliance

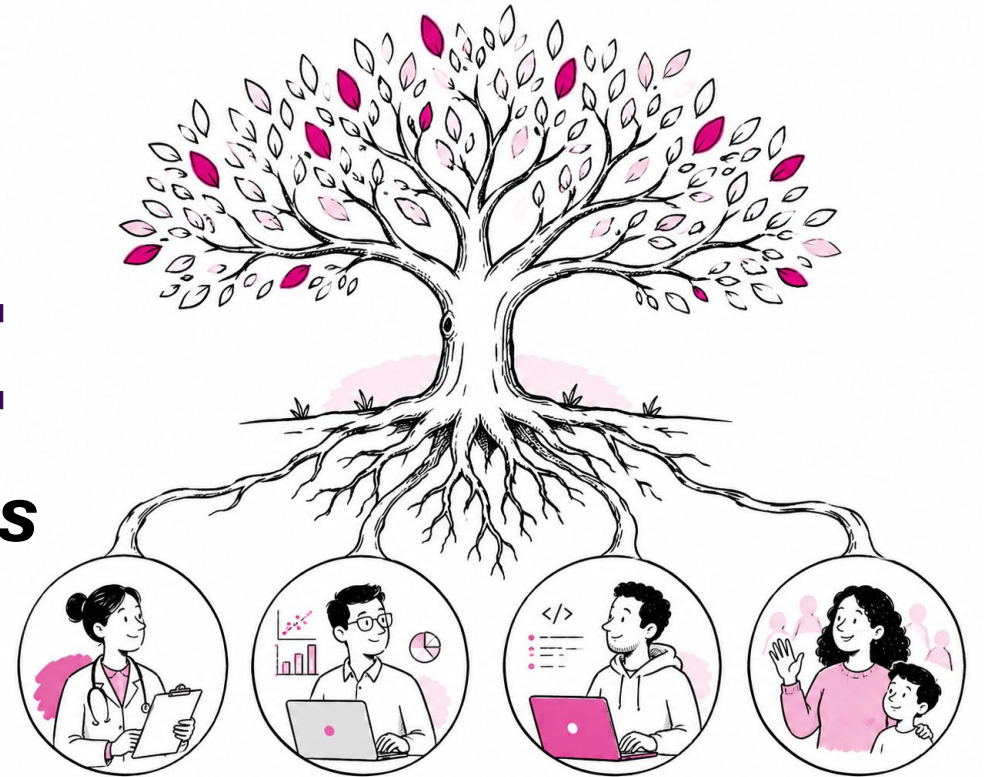


Eva Caamaño Gutiérrez

BIOMEDASA Project Lead
and Director, Computational Biology
Facility (University of Liverpool)

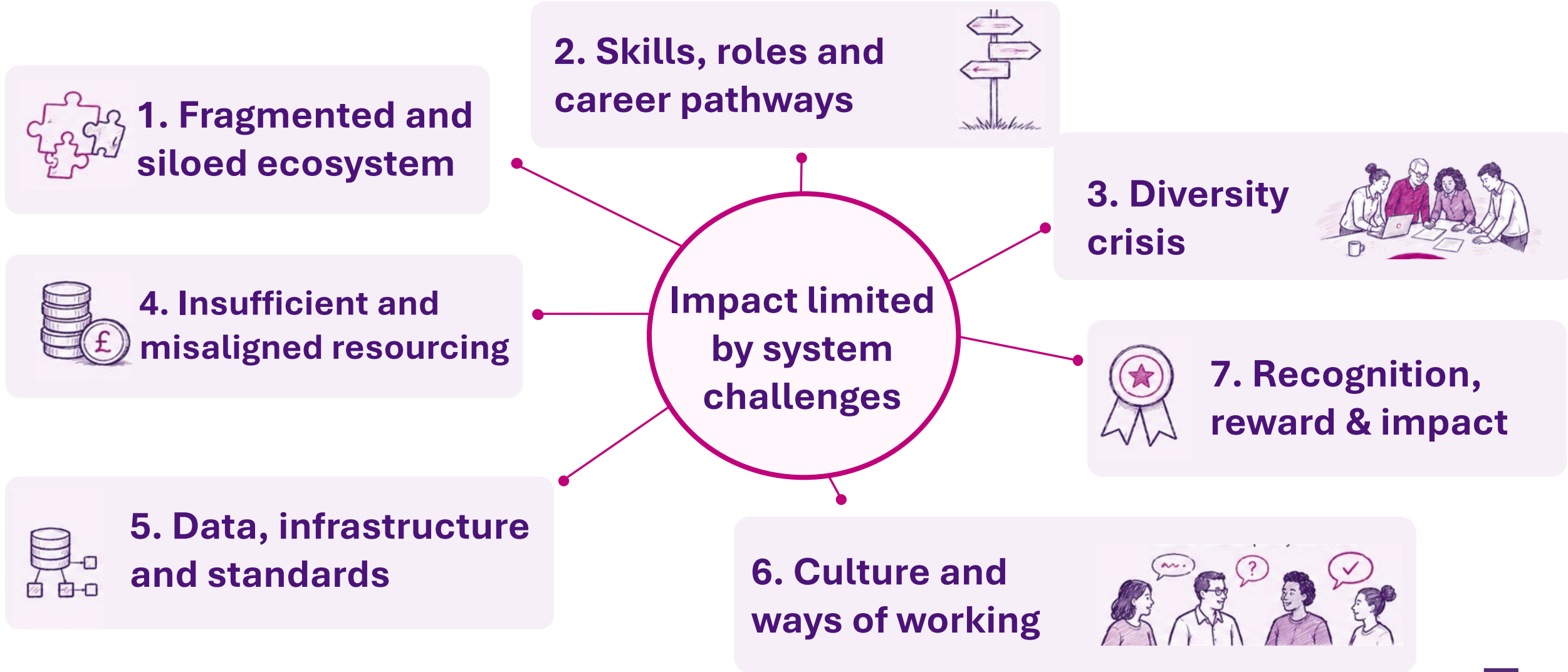
BETTER TEAMS, BETTER SCIENCE

*Cross-award evidence synthesis
from the MRC-BDSLA:
Emerging insights*



CHALLENGES IN BIOMEDICAL DATA SCIENCE

Insights inspired by the MRC report *The Opportunity of biomedical data science* (2022)





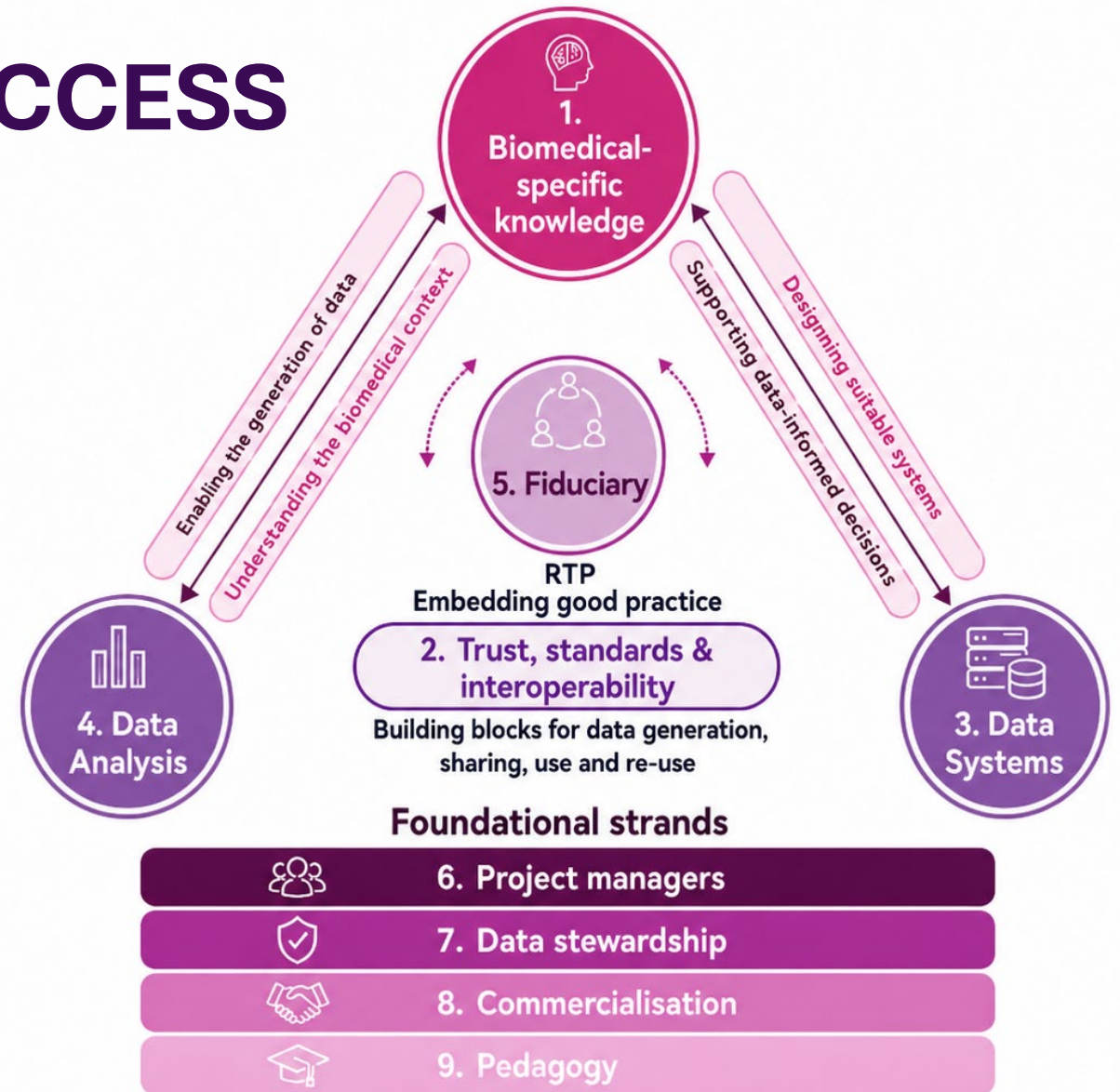
**Biomedical data science depends on
connected people, data and
infrastructure**

A CLEAR RECIPE FOR SUCCESS

Multi-disciplinarity

Inter-disciplinarity

Adapted from *The opportunity for biomedical data science, MRC 2022*





**MRC launched the Biomedical Data
Science Leadership Awards in response**

That's us!



BIOMEDICAL DATA SCIENCE LEADERSHIP AWARDS



ABDC

Advancing Biomedical Data Science Careers

✉ abdc@turing.ac.uk



BIOMEDASA

Biomedical Data Science Accelerator

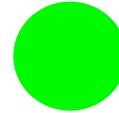
✉ biomedasa@liverpool.ac.uk



HxC

Healthier Science through Collaboration

✉ hxc@ukdri.ac.uk

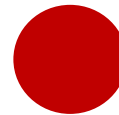


INTEGRATE

INTEGRATE

Embedding a supportive culture for data science in biomedical research

✉ integrate@imm.ox.ac.uk

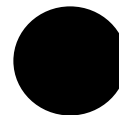


PROMOTE

PROMOTE

Progressing routes and opportunities through mentoring, openness, training and equity

✉ d.r.westhead@leeds.ac.uk



THE FIVE KEY TAKE AWAYS

Culture, integrity and team design improve scientific outcomes

- 1** Interdisciplinary research is often transformative yet is under-exploited and under-valued by institutions and funders alike.
- 2** Data expertise is essential but often an after-thought.
- 3** Barriers to change span from early to higher education, and block cross-sector career transitions.
- 4** Inequity of reward and recognition of data professionals, further exacerbated by other EDI considerations.
- 5** GenAI is major disruptor compounding existing challenges and accelerating the need for expert teams.

Interdisciplinary science is under-resourced

Interdisciplinary research can be more productive and rewarding, but it imposes additional relational and organisational work that institutions rarely resource explicitly.

93%

Advances
research

78%

More
satisfying

yet

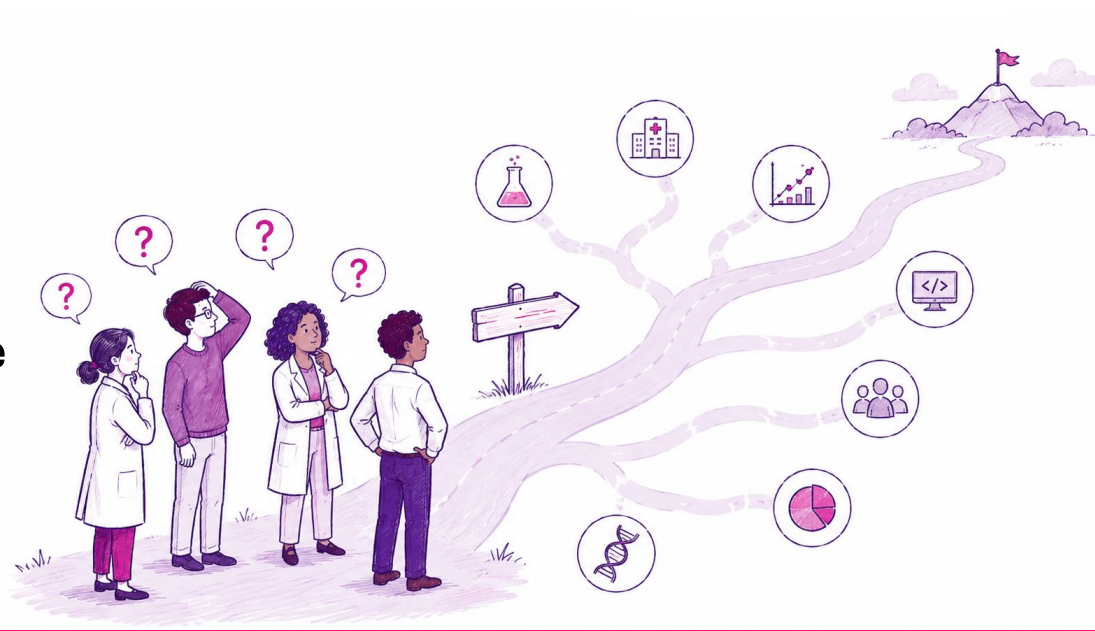
70%

Team-work
is harder

60%

Careers are
harder

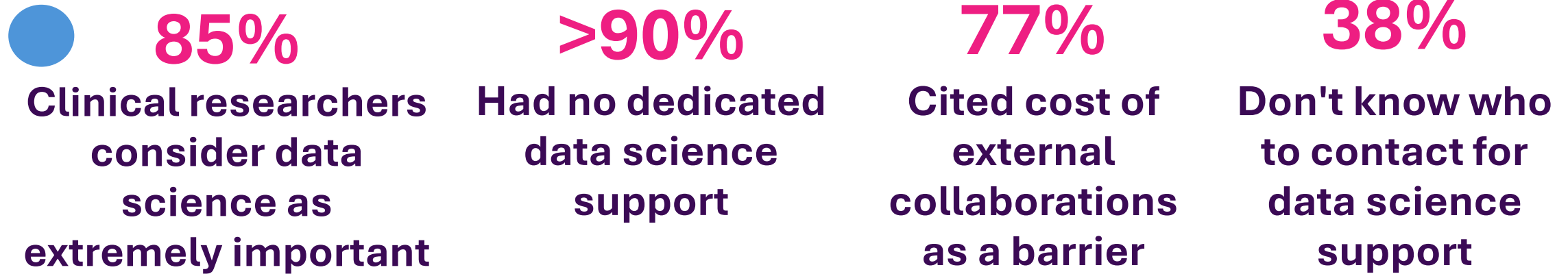
Inter and  multidisciplinary are not well funded



“data scientists, and support staff are often excluded from study design discussions, authorship decisions, and strategic planning, even when their input is critical to the success of projects.”

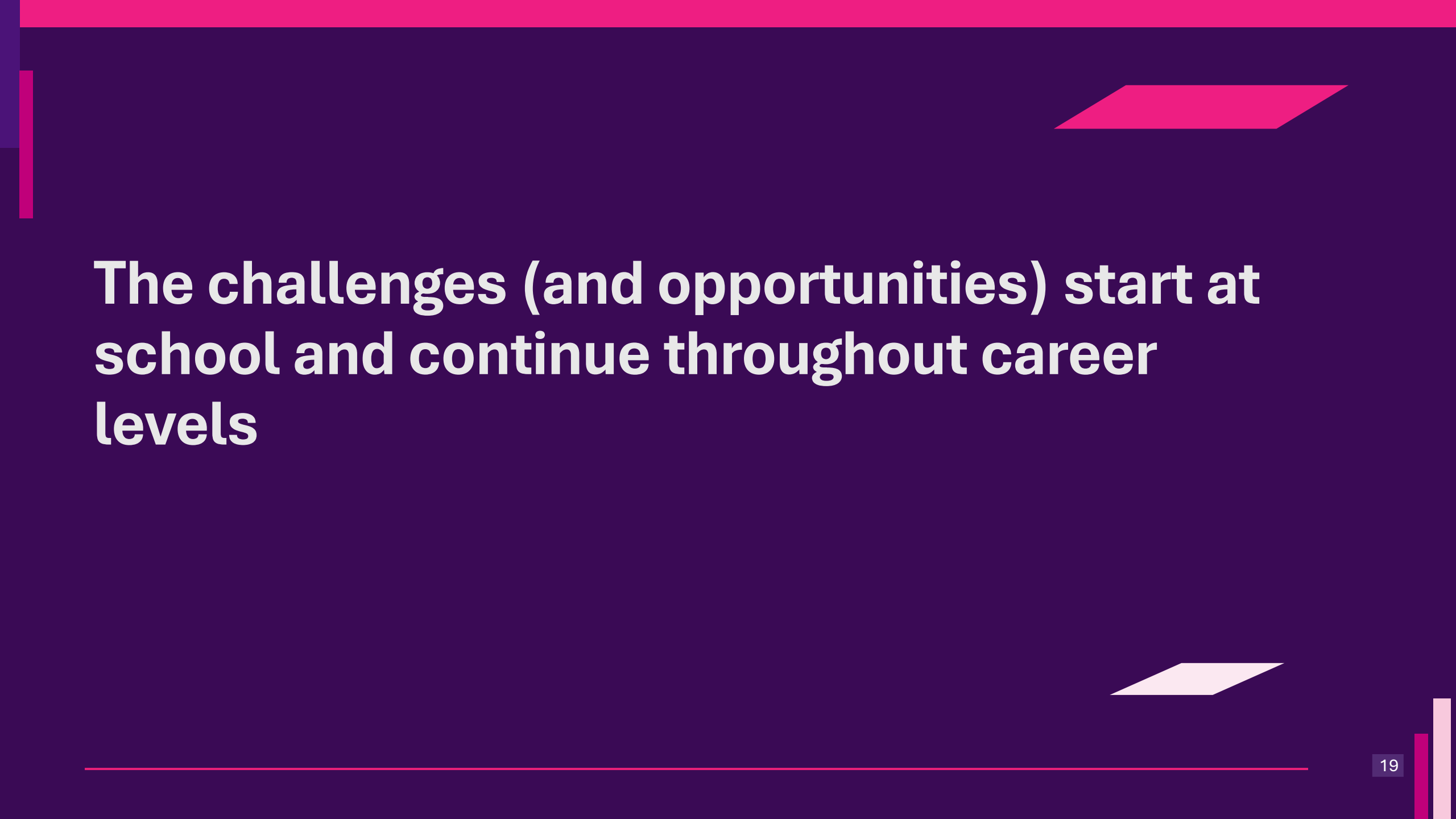
Data expertise is central – but still added too late

Data experts are frequently asked to rescue decisions they were not involved in making



But who does the data work?

- No complete set of competency frameworks for biomedical data science roles
- No clear career pathways
- Currently, difficult to understand roles needed in teams

The slide features a dark purple background with several pink geometric shapes: a vertical bar on the left, a parallelogram in the top right, and another parallelogram in the bottom right. A thin pink horizontal line is located near the bottom of the slide.

The challenges (and opportunities) start at school and continue throughout career levels

At the start of our formative years



01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science & Research culture

1:4

Ratio of girls to boys choosing Computing GCSE

3/4

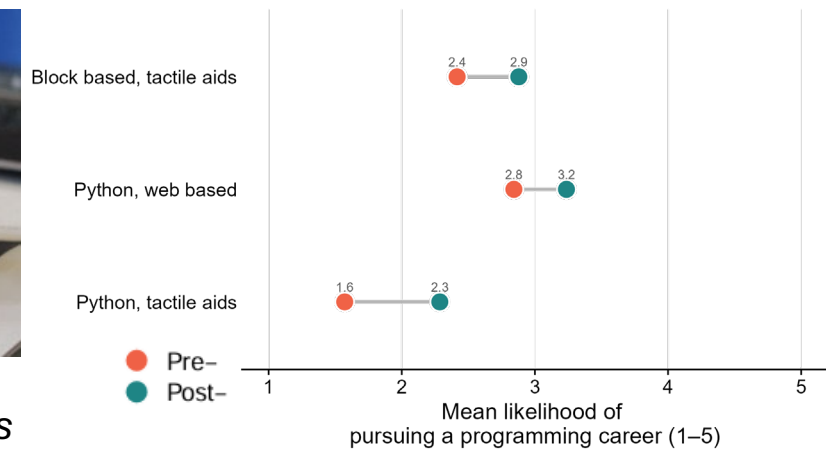
Lack of suitable computer rooms and other IT were a barrier

"GCSE students cannot use a keyboard and mouse because they have only used touch screens"

Children are more confident after interventions



"I really enjoyed this section, and I hope we do this again next year"



"I think this was a great opportunity for us to understand and learn more on how to code"



At the HE specialisation: MSc



35%

Students had limited ML/AI training access

35%

Lack opportunities for networking

31%

Limited teamwork experiences

They miss more practical learning
less “notebook following”

01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science & Research culture

MScs are an important entry point to BMDS careers



52%

Biomedical data scientists stated MSc as interdisciplinary entry point



Biomedical data professionals commonly bridge specialist domains and move between lab, computational and organisational cultures. These bridging skills are usually learnt through experience rather than formal training



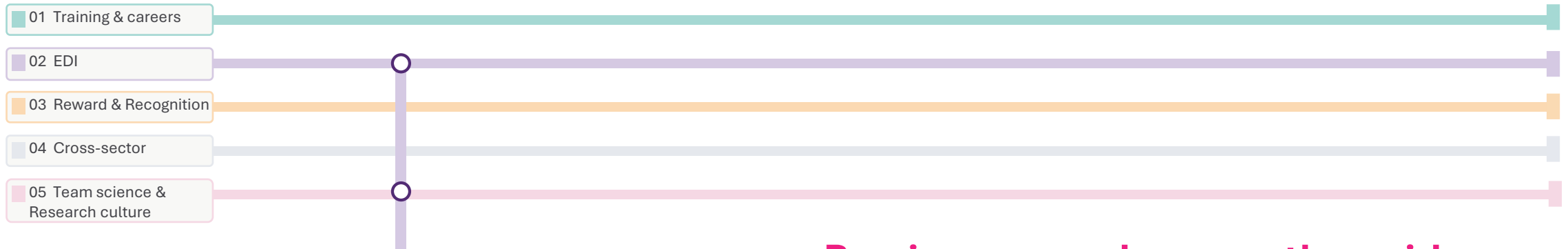
Researchers confirm team building opportunities happen in informal settings only



Early- mid career stage

Opportunity, voice and credit are not distributed equally

Inclusive teams require inclusive design, not only inclusive intentions.



Workplace EDI problems reported by university respondents



31.6%

**Female
respondents**

5.1%

**Male
respondents**



Barriers named across the evidence

- Caring responsibilities
- Short-term contracts
- Unequal admin load
- Flexible-working uncertainty
- Non-linear career barriers

Talent is widely distributed. Access to opportunity, recognition and participation is not.

But we know best teams are diverse

PNAS RESEARCH ARTICLE | SOCIAL SCIENCES  OPEN ACCESS 

Gender-diverse teams produce more novel and higher-impact scientific ideas

Yang Yang^{a,b,c}, Tanya Y. Tian^d, Teresa K. Woodruff^e, Benjamin F. Jones^{f,g}, and Brian Uzzi^{h,i,j,1}

Edited by Susan Fiske, Princeton University, Princeton, NJ; received January 16, 2022; accepted July 24, 2022

DOI: 10.1038/s41467-018-07634-8 OPEN

The preeminence of ethnic diversity in scientific collaboration

Bedoor K. AlShebli¹, Talal Rahwan^{1,2} & Wei Lee Woon^{1,3}

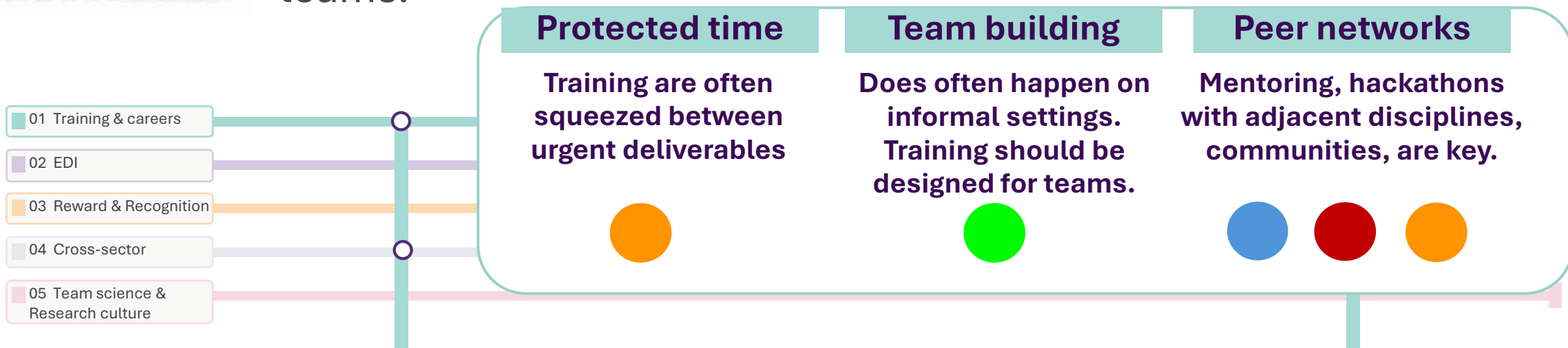
Inspired by the social and economic benefits of diversity, we analyze over 9 million papers and 6 million scientists to study the relationship between research impact and five classes of diversity: ethnicity, discipline, gender, affiliation, and academic age. Using randomized baseline models, we establish the presence of homophily in ethnicity, gender and affiliation. We then study the effect of diversity on scientific impact, as reflected in citations. Remarkably, of the classes considered, ethnic diversity had the strongest correlation with scientific impact. To further isolate the effects of ethnic diversity, we used randomized baseline models and again found a clear link between diversity and impact. To further support these findings, we use coarsened exact matching to compare the scientific impact of ethnically diverse papers and scientists with closely-matched control groups. Here, we find that **ethnic diversity resulted in an impact gain of 10.63% for papers, and 47.67% for scientists.**



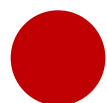
Career wide problems

The skills gap is also a team-design gap

Training matters, but individuals cannot compensate for under-designed teams.



Biomedical data professionals find



65% Barriers to accessing training

77% Want career workshops

68% Want wider peer networks

51% Want industry secondments

“Development is rarely built in for the data scientists, which is odd, because of all people we need to keep up to date with rapidly evolving methods”



Career wide problems

Reward and recognition

Progression is difficult to frame in the current culture

Recognition fades with distance from the work



>85%

Valued by immediate
teams, managers and
collaborators

~57%

Valued by departments
or the wider
biomedical community

01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science &
Research culture



60%

No clear and
achievable routes
to progression



43%

Limited career
pathways for
progression

*'It is difficult for data scientists to secure
external funding in this role.'*

Work performed

research · software · methods · datasets · documentation ·
training · infrastructure · outreach

Work rewarded

publications · grants





Career wide problems

Reward and recognition

Unequitable recognition across the various disciplines

“Despite doing all data analyses, figures, writing of methods, results and some of discussion there is still question about authorship”

01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science & Research culture

“The health data scientists are fragmented across clinical areas, with little opportunity to collaborate or support for their development. They are treated as service staff, not independent scientists with their own ambitions.”

“How line managers perceive their statisticians or data scientists compared to their clinical fellows is very different. The data scientists and statisticians always end up being spread so thin [...] you just get sent all over. And then the clinical fellows aren’t, they are asked ‘Do you want to now apply for the NIHR fellowship?’”

“[there is] no career progression in my role”



Career wide problems

Cross-sector porosity – perceptions and experiences tell us is not yet working

01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science & Research culture

"Now I have been in industry for a few years, moving back to academia seems difficult (if not impossible)...I truly believe my few years in industry has given me a lot of experience that would still be relevant in academia"

"I joined academia after working as a medical statistician in industry for over 10 years ...Not having a PhD has been a barrier to career progression."

"Colleagues that have gone into industry have generally positive experiences, but also the colleagues that went into the industry are less, I think, creative than I am."

"But there is so much to industry than, you know, just doing these boring things. [...] And, they just can be equally exciting. So, I think just lack of understanding that industry could be actually exciting."



Career wide problems

Interdisciplinarity needs to be designed

"I feel most people are not trained in these basics and know how to analyse omics data well which is scary as it is becoming expected of us to be able to do all of these multiple techniques and types of analysis in our research"

01 Training & careers

02 EDI

03 Reward & Recognition

04 Cross-sector

05 Team science & Research culture

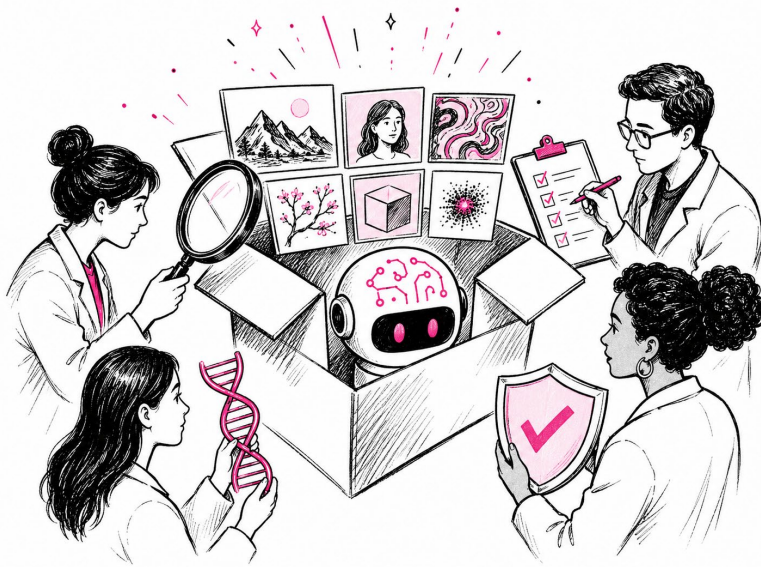
"There are things now that people are trying to create in our field, like a career ladder that you can climb up. At the moment, it doesn't feel like one."

"This wouldn't have been a career option on my radar [...] it's not hidden, it's just not at the forefront of people's minds."

"The iterative process in data science is cognitively demanding and not always recognised, with some still perceiving it as just clicking buttons."

GenAI disrupting research practices and teams

Skills, practices and responsibilities in research are changing



New **skills** → what are they and who needs them?



New **roles** → AI specialists or part of existing roles?



New **team** practices → GenAI and Agentic AI may bring different types of collaboration



New **tasks** and accountability → Responsible AI

"I'm a little bit afraid that people are just using tools without understanding them, which risks misunderstanding biological results."



"I'm not entirely sure that people in charge are clear about what to do with AI."



"It was very discouraging to turn in a report [...] and receive a lower grade than others who turned in a ChatGPT report."

THE FIVE KEY TAKE AWAYS

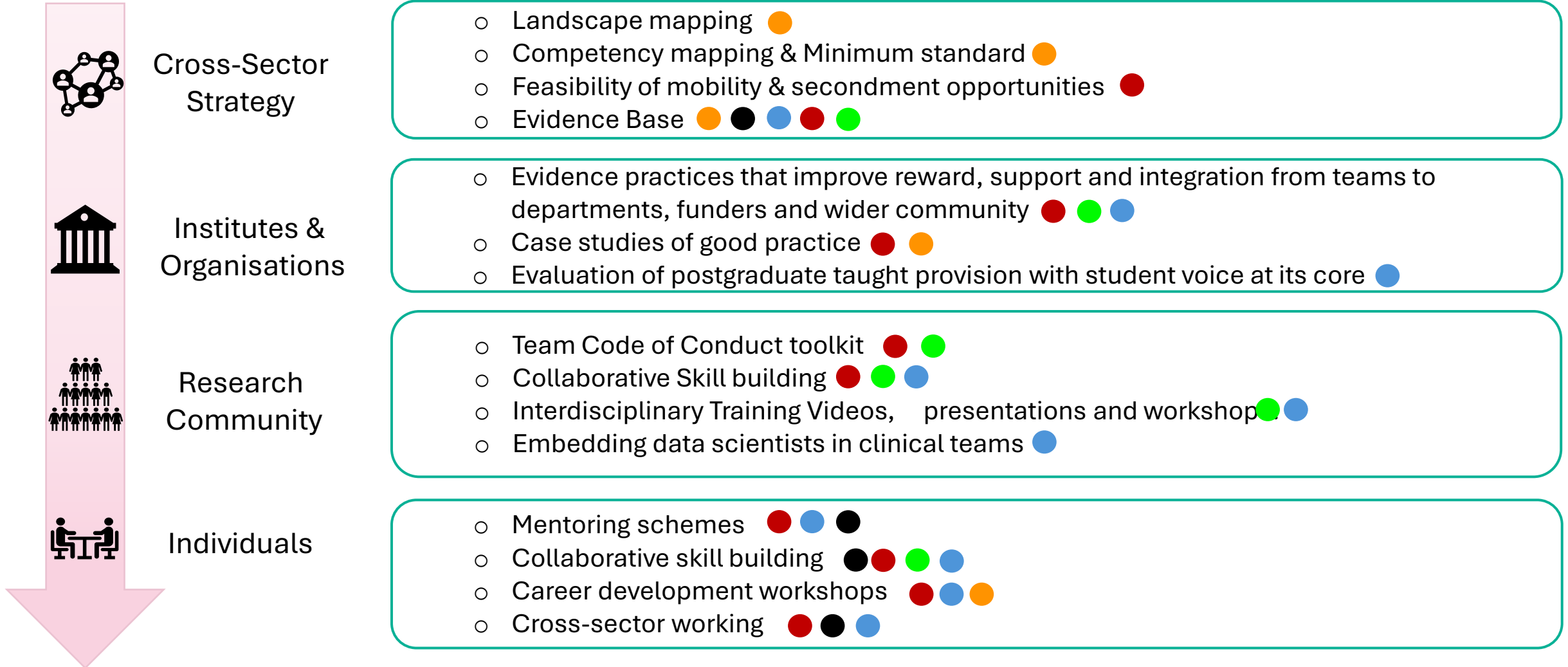
Culture, integrity and team design improve scientific outcomes

- 1** Interdisciplinary research is often transformative yet is under-exploited and under-valued by institutions and funders alike.
- 2** Data expertise is essential but often an after-thought.
- 3** Barriers to change span from early to higher education, and block cross-sector career transitions.
- 4** Inequity of reward and recognition of data professionals, further exacerbated by other EDI considerations.
- 5** GenAI is major disruptor compounding existing challenges and accelerating the need for expert teams.

But, are there solutions that can work?



BDSLA evidencing solutions for impact across R&I ecosystem



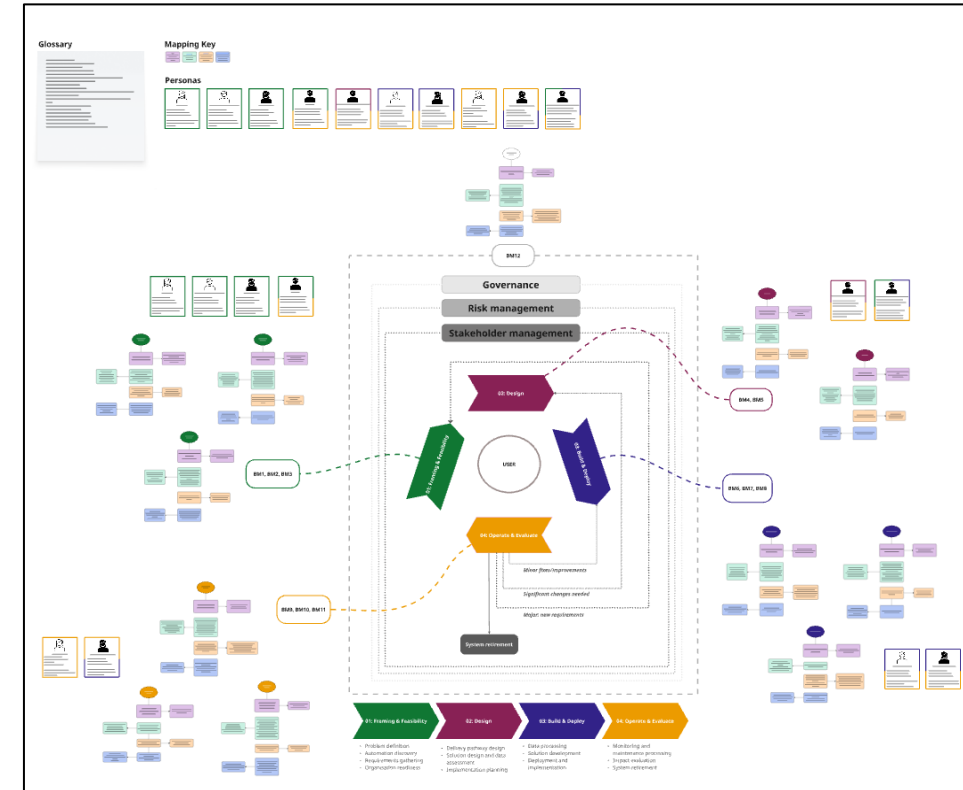
Competency mapping

Value starts with definitions

- Benchmarking and mapping of existing trusted standards to **define a "Minimum Standard"**
- The **three pillars** of our mapping:
 - AI Skills for Business (AISfB)
 - NHS National Competency Framework (V2)
 - ISCB (International Society for Computational Biology) V3

Outcomes

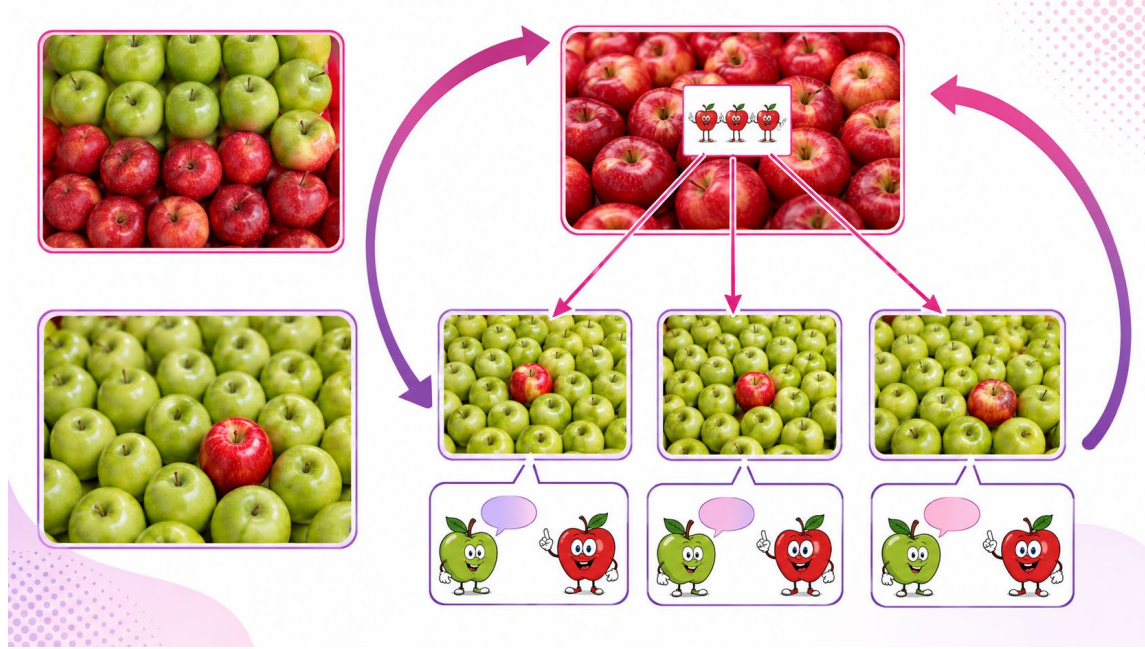
- **12 minimum standards**, 4 proto-personas, 10 specific personas
- **Ready for use** by biomedical organisations as **starting point** for own competency frameworks
- **Democratised access** – help with understanding of skills gaps and potential career choices and paths



Open access
through EMBL-EBI
Competency hub

Better team design in cost-effective intervention

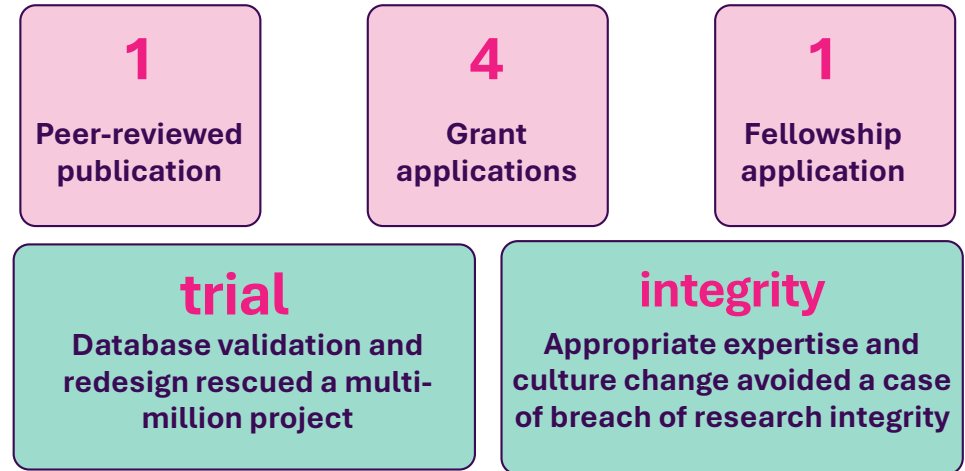
Value and trust flourish with sustained collaboration



Provide Purpose and motivation

"It was a good opportunity to support a research team that is working on improving the quality of outcomes for patients.[...] I felt like I had could use my skills and contribute to supporting a good cause".

Unlocks outputs and quality



Sow the building blocks of interdisciplinarity

"I found it challenging to understand what certain variables in the dataset mean, and primarily how to treat/use them for analysis. This required a fair amount of back and forth from my side to understand what the data means".

"Communication is key. Ask lots of questions, you'll learn about the research area and feel confident when contributing to the team".

Evidence for interdisciplinary research success

Mapping key drivers to make data science researchers feel valued, supported and integrated within their teams, institutes and funders



What training, initiatives or opportunities (formal or informal) have most helped you to progress in your personal and professional development and why?

General

Informal team meetings
Network
Community
Openness
Rigor
recognition
interdisciplinary supervision

Mentoring

Post-doctoral mentoring programme with senior mentor from a different field (Cambridge)
NERD Mentorship Programme - Glasgow
Fellowship writing & network with other fellows/group leaders
Informal mentorship from colleagues
RSE Society Mentoring Programme
OLS Mentoring
ELIXIR Mentorship Programme

Personal Development workshops and training

Cambridge C2D3 (HPC) workshops and funding
EMBO lab management & leadership course
Communication skills
Leadership Training - University of Oxford
Management & leadership training
WIMM Leadership programme
SSI Fellow mentorship programme
Babraham mentorship programme
ARVO mentorship programme
NHS Graduate Management Training coaching
management & leadership training - the effective statistician Dr Alexander Schacht
Grant writing - MSD Oxford
RisingWise - University of Oxford / Cambridge enterprise programme
Interdisciplinary research training (University of Edinburgh, Turing Institute)
Research excellence training
Sports Coaching / Training
University of Manchester Disabled Leadership Programme

Development opportunities

Teaching training - e.g. Cambridge PGCTLHE PGCert training
Supervision and management opportunities

Committee membership

Event organization
Collaborative grant writing
Bioconductor community Champions
ARVO science communication training fellowship
Grant Panel observation
ECR Support Glasgow |
Development opportunities outside of primary role
ONS safe researcher training and accreditation
Within lab training on peer-reviewing, public speaking
PhD by Publication

Technical Training

Deep technical training in biology
PhD Statistics
Vocational training via Scientist Training Program & workshops
MSc Health data analytics UCL
NIHR predoctoral research Programme on-the-job training
RSE Training - Oxford RSE Group
BRDX courses - code
Cambridge Bioinformatics Courses
Importance of Experienced team
Working in industry
Online training courses - Coursera, MIT
EBI workshops
Hackathons
Project-based learning
MRC CGAT Fellowship
software carpentries
Capital Workshops / Courses
MRI imaging
AWS Training
MSc Computer Sciences CPD
Open University Mathematics Degree
Healthcare Management Postgraduate diploma

Conferences & Communities

GEOMED, METMA, RSS, ISEE
Bioconductor
Software Sustainability Institute (SSI)
Festival of Genomics
Women in Biotech events
BIA
Open science community
Software Carpentries
Biostars
stack overflow
Github
Turing Way
RSE initiatives
ELEAD
Oracle User Group
Open-Source Conferences

Funding opportunities

SSI Fellowship
Returning Carer Funding - Academic Development Fund
University of Oxford
Data Science Training Bursary R
NIHR Studentship for MSc funding

Internships / Secondments

WHO Internship
Industry Internship in Software development department
1 year secondment
3-month secondment
Lab visits
Undergraduate placements (one academic, one industry)
6-month secondment

Collaboration opportunities

3-day workshop with lots of unstructured time leading to collaborations with software engineers in the same niche field (systems neuroscience)
Hackathons

"informal mentoring by committed collaborators has been indispensable. Having a senior colleague who really cares about your career development is irreplaceable."

"Hackathons - have been great opportunities to meet other RSE's and people that are much better at coding than I am, so have gained experience from these people."

"I believe it was being curious and learning on my own, with projects and online courses (e.g. Coursera). I feel most workshops are way too short to teach you things in-depth...learning "hard skills" (new methods, techniques, programming languages) on my own is what made the difference most of the time."

What opportunities specifically from funders have been most helpful in your career development?

Panel observation, roundtables, symposium training

Widening criteria (RTPs, non-PhD, non-PIs.)

Structured project/fellowship funding across career stages

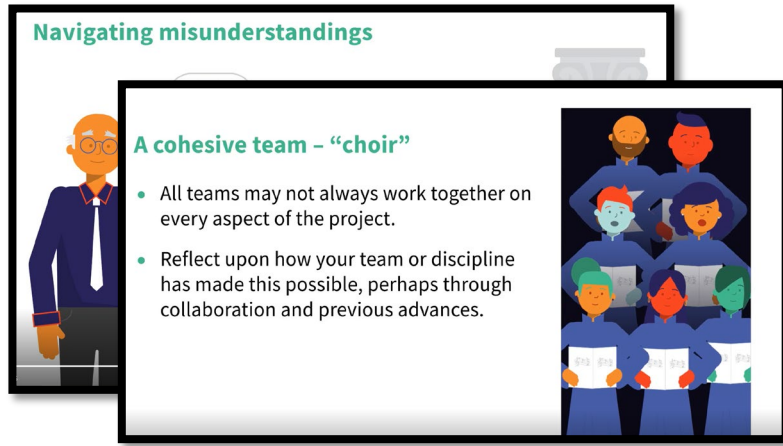
Funding for travel, HPC, Open Science

'Cheatsheets' of Training, initiatives, communities and opportunities that have supported professional development & progression

Evidence for interdisciplinary research success

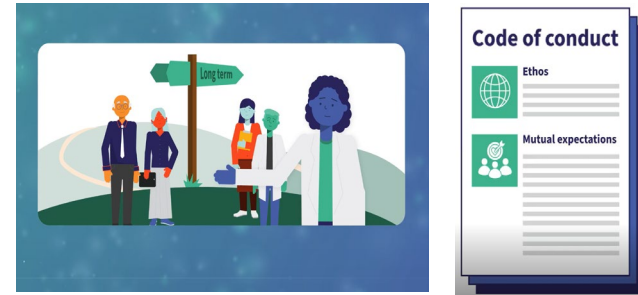
Finding what supports individuals and interdisciplinary teams in academic environments

CPD Accredited Bitesize Video Series



- 11-course video **interdisciplinary training** programme
- **Evidence-informed** topics identified through stakeholder survey
- **Co-produced** with ECRs, industry, patients, and public

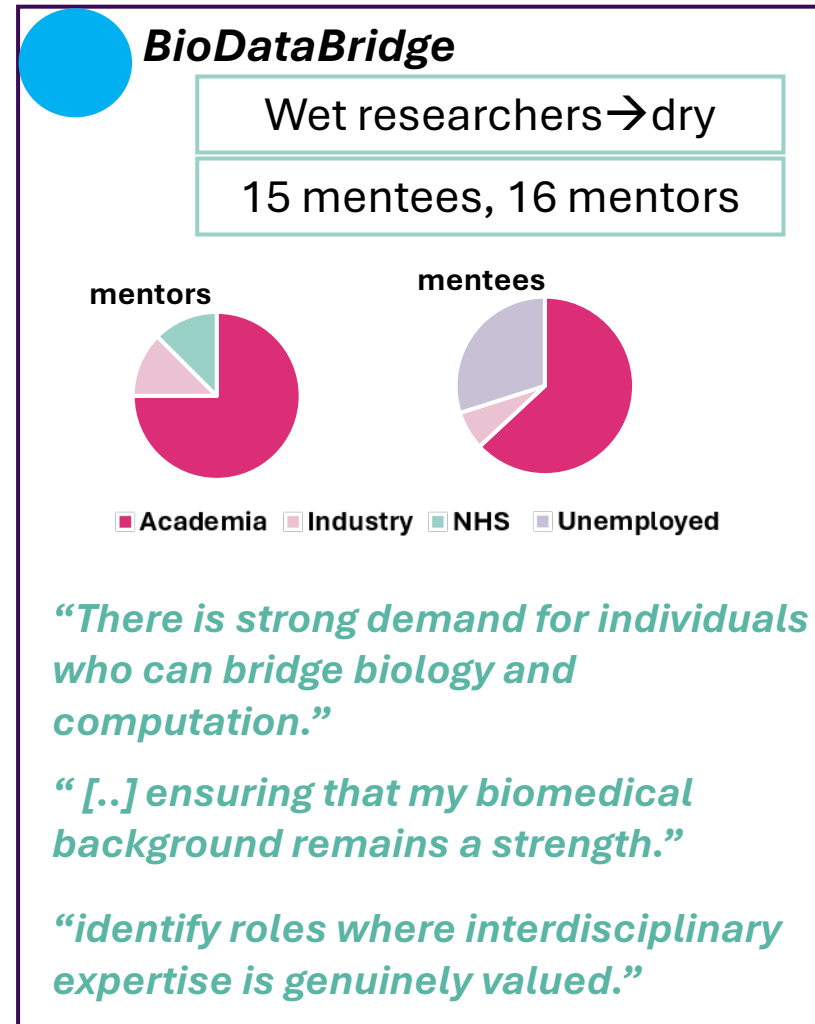
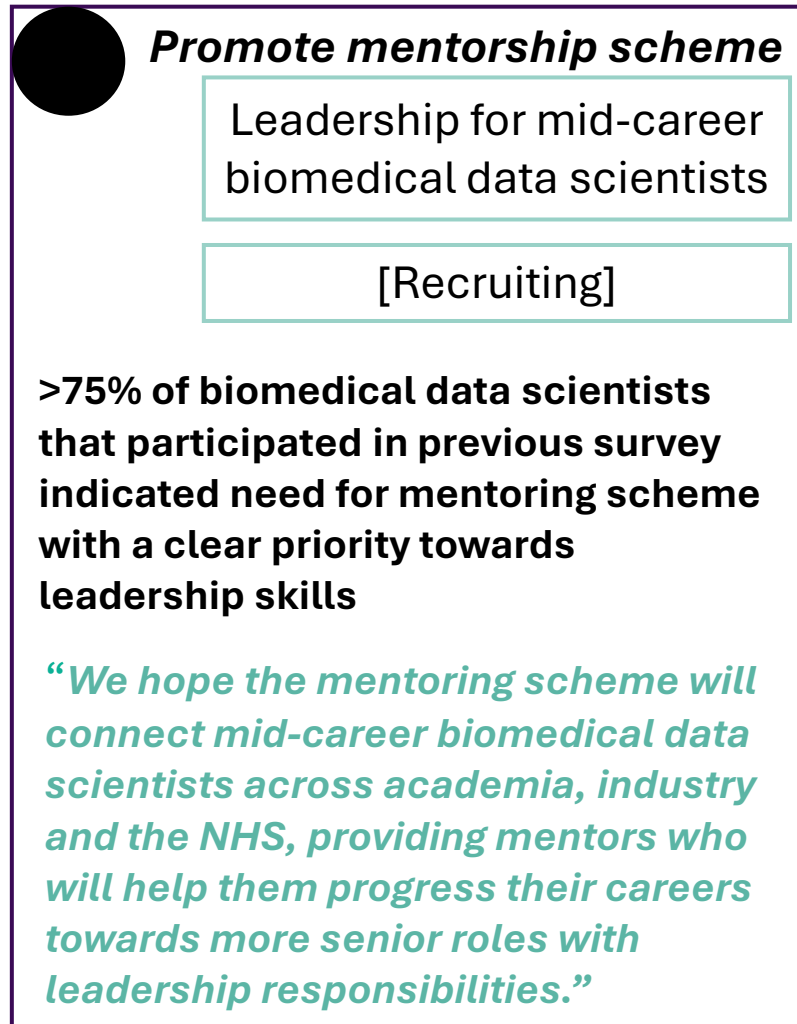
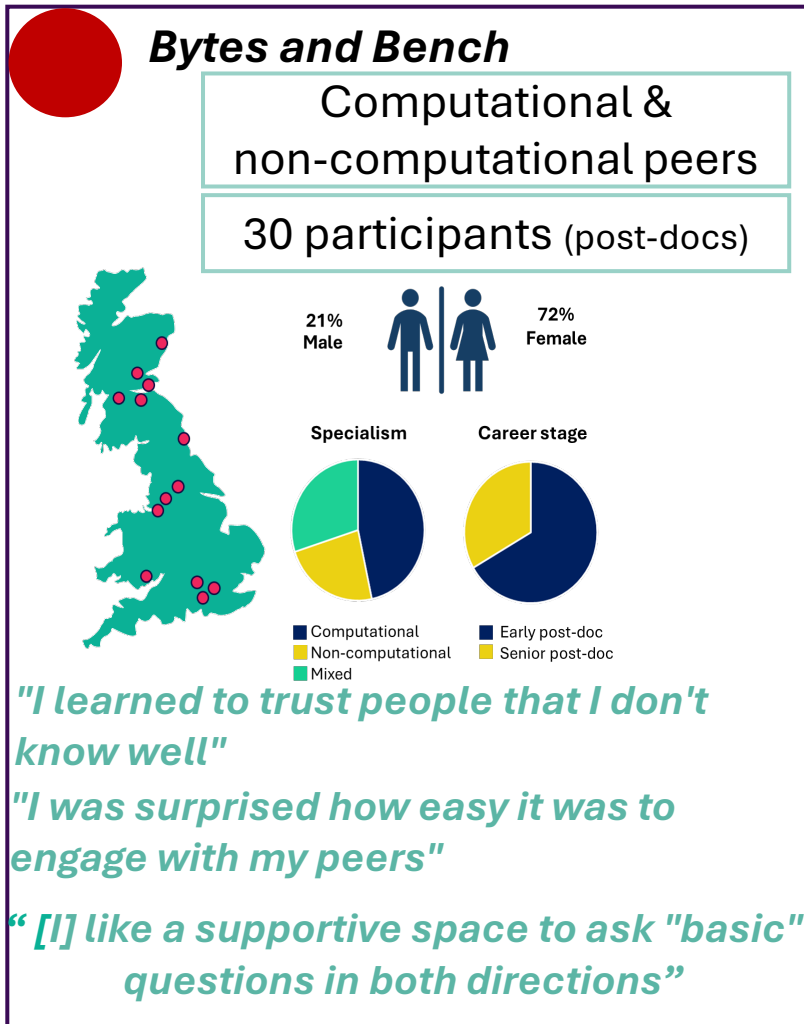
Team Code of Conduct Toolkit



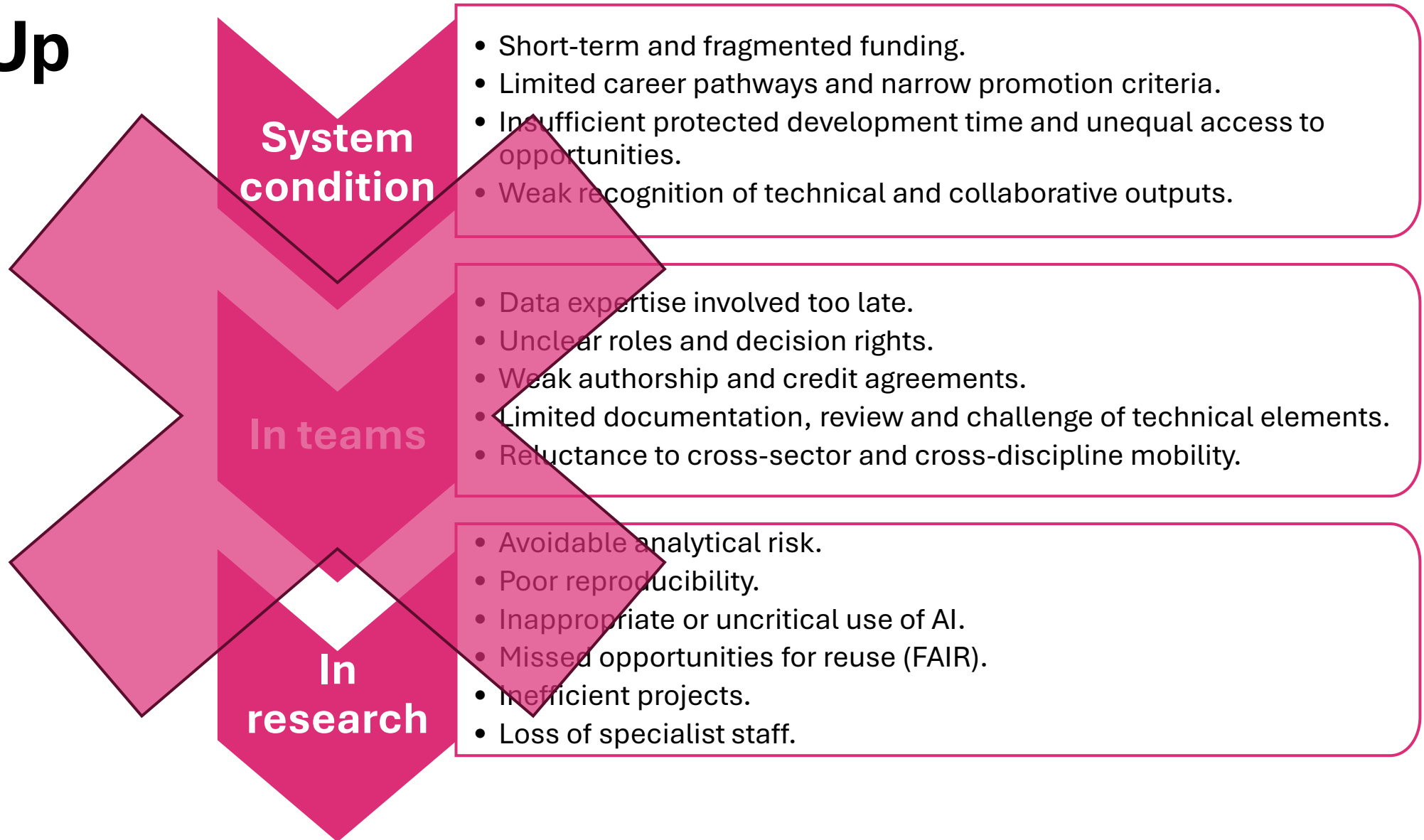
- Activities improved understanding of colleagues: ***"it helps you understand how your colleagues think."***
- Many felt activities would: ***"definitely help team dynamics"*** if completed by most of the team.
- **Leadership was seen as important to drive interest** in training and activities

Mentoring as a conduit for interdisciplinarity

Three schemes, high demand, multiple sectors



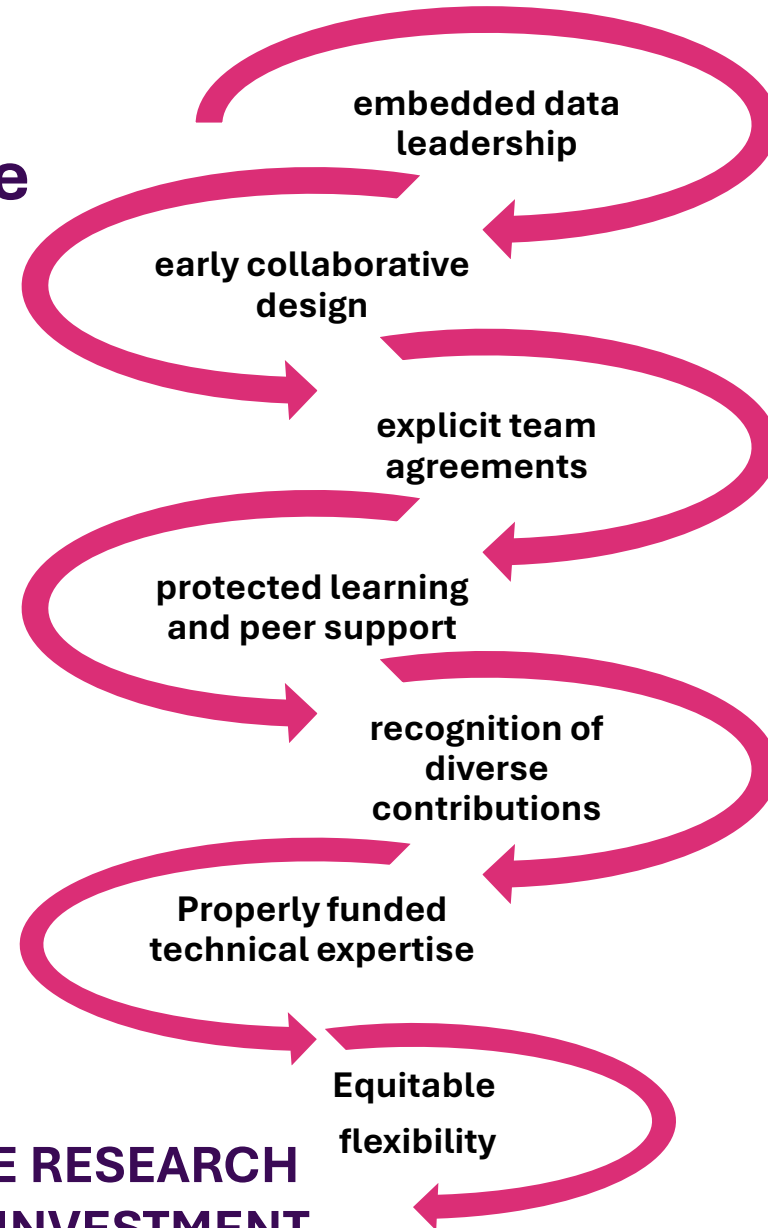
To Sum Up



LESS INNOVATIVE AND LESS INCLUSIVE RESEARCH, LESS VALUE PER INVESTMENT

Better teams are our key

Culture, careers and recognition are not separate from the science



**MORE INNOVATIVE AND INCLUSIVE RESEARCH
AND MORE VALUE PER INVESTMENT**



**Better teams are not
a cultural extra.**

They are infrastructure for research quality.





How do we get this body of evidence to make a difference?

How can we build on the progress already made and move towards systematic interventions?

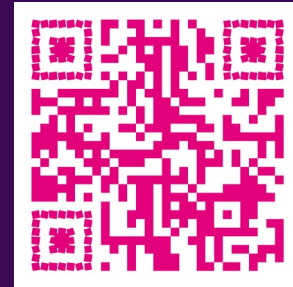
We can only make change together

Help us bring Evidence to Ecosystem



Many thanks

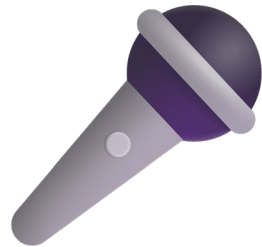
Stay in touch



Q&A

In room:

Hands up and microphone



All:

www.slido.com

#4049254





Future Directions and Drivers for Change in Biomedical Data Science and Team Research

Panel discussion



Speakers



Tony Burdett
Director, BioFAIR



Sayoni Das
SVP Bioinformatics,
PrecisionLife



Debbie Johnson
Head of Innovation,
Talent & Skills,
Innovate UK



Joanna Robinson
Director of Research Talent,
Skills and Careers,
Medical Research Council

Panel chair

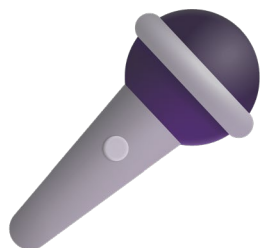


Tanita Casci,
Director, Research Strategy &
Policy Unit, University of Oxford

Q&A

In room:

Hands up and microphone



All:

www.slido.com

#4049254



Coffee break!

Please be back by 11:35

...and sit at the table with your name



Making change happen

Part 1: Visioning and current activities



Denise Bianco

Senior Research Community Manager,
ABDC Project,
The Alan Turing Institute

Part 1: The roots and the blue sky (60 min)

Objective: Surface what we are already doing and how it connects to the overarching vision

Question 1: The blue sky (20 min)

- *What does this vision represent for you, and what's the ideal scenario you'd like to see?*
- Write your answer(s) on the **blue sticky notes** and position them on the tree leaves

Part 1: The roots and the blue sky (60 min)

Objective: Surface what we are already doing and how it connects to the overarching vision

Question 2: The roots (40 min)

How does your work align with the vision, and what activities are you currently carrying out to support it?

- Write each activity on a separate post-it note
- Be specific and concise. Focus on concrete actions, not intentions
- Please add your name and/or organisation to the post-it note
- Place your post-its on the board under the relevant theme and share with the group
- Multiple post-its per person are encouraged
- Work collaboratively to cluster similar activities within each theme

What does this vision represent for you, and what's the ideal scenario you'd like to see?

5	We envision: Better Teams, Better Science, Better Health. Building a visible, fair, and impactful biomedical data science culture that gives funders, institutions, and researchers the evidence to drive real change across UK biomedical ecosystem.				
4	Equity, Diversity & Inclusion	Rewards & Recognition	Training & Careers	Cross-sector working	Team Science & Research Culture
3					
2	Output What How Who Indicators Outcome it contributes to	Output What How Who Indicators Outcome it contributes to	Output What How Who Indicators Outcome it contributes to	Output What How Who Indicators Outcome it contributes to	Output What How Who Indicators Outcome it contributes to
1					

How does your work align with the vision and what activities are you currently carrying out to support it?

Lunch break!

Please be back by 13:30





Making change happen

Part 2: Defining Actionable Steps



Part 2: Defining Actionable Steps (90 min)

Objective: Move from current activities (the “roots”) to concrete next steps towards the vision

Step 1: Define Output 1 (60 min)

- Decide with your group one or two pillars to focus on together (or more if time allows)
- The first ~10 min are for individual reflection
 - use sticky notes to write down your thoughts
 - Sharing with the group
- **Using the template card, collaboratively define Output 1:**
 - The **first concrete, specific step** towards the vision (this is your "1" on the scale)
 - Something realistic and actionable
 - You may:
 - Work on multiple pillars, or
 - Create multiple cards within the same pillar (extra cards are provided)

Output



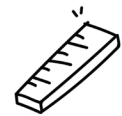
What



How



Who



Indicators



Outcome it contributes to

Part 2: Defining Actionable Steps (90 min)

Objective: Move from current activities (the “roots”) to concrete next steps towards the vision

Step 2: Sketch Output 2 (30 min)

- **Based on Output 1, briefly define Output 2 (the step immediately after Outcome 1)**
 - Keep this lighter and less detailed
 - Focus on direction rather than full planning
 - You may:
 - Work on multiple pillars, or
 - Create multiple cards within the same pillar (extra cards will be provided)

Output



Making change happen

Share out



Emma Karoune

Project Lead, ABDC Project,
The Alan Turing Institute

Reflection

Share out:

What is the first tangible step (Output 1) in your roadmap? What follows?

- Each group to share (2–3 min)

Reflection:

What is your one key takeaway from today, and what will be your next action?

- Slido
- Open discussion

www.slido.com

#4049254



Next steps

Workshop outputs

- Feedback form
- Workshop summary
- Action plan

Research Consulting

- Working with us on maximising our project outcomes and legacy
- Will be contacting stakeholders for:
 - Interviews
 - Groups

Date for your diary

- Final Event!
- Hosted at The Spine, Liverpool
- **3rd March 2027**
- Project final results, outcomes and next steps

Contact us!

Contact form



Thank you!

Thank you for attending and giving input on this important topic

Stakeholder Working Group

- Led organisation of this event

Executive Team

- Supported Working Group

Thanks to all **BDSLA project teams** for their input and help with this event:

- ABDC
- BIOMEDASA
- HxC
- INTEGRATE
- PROMOTE